KIMQ8550 Ubuntu 22.04 OS Update Guide

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General information

This guide describes how to reflash Ubuntu 22.04 OS version V3.0.0 onto the AIBOX device on the KIMQ8550 platform, using the QDL (Qualcomm Device Loader) tool in EDL mode. The host PC can run either Ubuntu or Windows.

Item	Value
Platform	KIMQ8550 (AIBOX)
Operating system	Ubuntu 22.04 — V3.0.0
Flashing tool	QDL (Qualcomm Device Loader)
Connection	USB Type-C cable (data-capable)
Default account	`root` / `oelinux123`

> **Warning:** Updating the OS will erase all data on the device. Please back up any necessary data before proceeding.

1. Preparation

Install the QDL tool and download the OS V3.0.0 package to the host PC before putting the device into EDL mode.

1.1. Install the QDL tool

1. Download the QDL tool from Qualcomm Software Center — Qualcomm Device Loader.
2. Follow the bundled **QDL_User_Guide.pdf** to install it for your environment (Windows or Ubuntu), including the USB driver if the guide requires one.
3. Verify the installation: open a terminal (Ubuntu) or Command Prompt/PowerShell (Windows) and run `qdl --help`. If the usage help is printed, the installation succeeded.

1.2. Download the Ubuntu 22.04 V3.0.0 OS package

1. Download the OS package from the Google Drive folder.

2. Extract it to a folder on the host PC. Use a short path without spaces or special characters, for example `~/kimq8550_os_v3.0.0` or `D:\kimq8550_os_v3.0.0`.
3. Check that the extracted folder contains: `xbl_s_devprg_ns.melf`, the `rawprogram*.xml` files, the `patch*.xml` files, and the OS image files.

2. Put the AIBOX into EDL mode

The AIBOX enters EDL mode when powered on while the reset button is held; the host PC recognizes it correctly when a Qualcomm USB device with ID **05c6:9008** appears.

2.1. Steps

1. Disconnect the power supply and all Type-C cables from the AIBOX.
2. Press and hold the reset button on the AIBOX, then connect power.
3. Keep holding the reset button for about 5–7 seconds, then release it. The AIBOX is now booted into EDL mode.
4. Connect the AIBOX to the host PC with a Type-C cable.

2.2. Check on Ubuntu

Run:

```
lsusb
```

Detected OK when a line contains ID `05c6:9008`, for example:

```
Bus 001 Device 012: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL
mode)
```

The Bus/Device numbers may differ; only the ID `05c6:9008` matters. To filter quickly: `lsusb | grep 05c6:9008`.

2.3. Check on Windows

Open **Device Manager** (`devmgmt.msc`).

Detected OK when a device appears without a yellow exclamation mark, typically **Qualcomm HS-USB QDLoader 9008 (COMxx)** under *Ports (COM & LPT)*. The displayed name may vary depending on the driver installed per QDL_User_Guide.pdf. To confirm: right-click the device → *Properties* → *Details* tab → *Hardware Ids*; the value must contain `VID_05C6&PID_9008`.

Not OK in the following cases:

- No Qualcomm device is listed: the AIBOX is not in EDL mode, or the Type-C cable is charge-only.
- The device shows a yellow exclamation mark (e.g. *QUSB_BULK* under *Other devices*): the USB driver is not installed.
- The device has a PID other than `9008`: the AIBOX booted normally instead of into EDL. Repeat section 2.1.

3. Flash the OS to the AIBOX

Open a terminal in the OS V3.0.0 folder extracted in section 1.2, then run the flashing command for your host OS. Do not unplug the cable or power off the AIBOX during flashing.

3.1. On Ubuntu

```
cd ~/kimq8550_os_v3.0.0
qdl xbl_s_devprg_ns.melf rawprogram*.xml patch*.xml
```

If you get a USB permission error (permission denied), rerun the command with `sudo` or configure a udev rule as described in QDL_User_Guide.pdf.

3.2. On Windows

Command Prompt on Windows does not expand the `*` wildcard, so you must enter the exact list of xml files.

1. List the xml file names in the folder:

   ```
   cd /d D:\kimq8550_os_v3.0.0
   dir /b rawprogram*.xml patch*.xml
   ```

2. Run the flashing command with every file name listed above. Example (the actual list must match the output of the `dir` command):

   ```
   qdl xbl_s_devprg_ns.melf rawprogram0.xml rawprogram1.xml rawprogram2.xml
   rawprogram3.xml rawprogram4.xml rawprogram5.xml patch0.xml patch1.xml
   patch2.xml patch3.xml patch4.xml patch5.xml
   ```

3.3. How to tell flashing succeeded

- QDL processes the whole file list without errors and returns to the command prompt.
- The AIBOX reboots automatically; the `05c6:9008` device disappears from `lsusb` / Device Manager.

4. Boot and log in

version

After flashing completes, the AIBOX boots into the new OS automatically. The first boot may take longer than usual.

1. Log in with user `root`, password `oelinux123`.
2. Check the OS version:

   ```
   cat /etc/os-version
   ```

   The result is correct when the `VERSION` line shows Ubuntu 22.04.
3. It is recommended to change the default password with `passwd` if the device will be connected to a network or handed over to end users.
4. Restore your backed-up data (if any).

5. Troubleshooting

Symptom	Common cause	Fix
Host PC does not see device `05c6:9008`	AIBOX not in EDL mode; charge-only Type-C cable	Repeat section 2.1 and hold reset for the full 5–7 seconds; use a data-capable cable or another USB port
Windows shows the device with a yellow exclamation mark	USB driver not installed	Install the driver per QDL_User_Guide.pdf, then unplug and replug the cable
Ubuntu reports a USB permission error	Current user lacks access to the USB device	Run the command with `sudo` or configure a udev rule
QDL cannot find the xml files	Terminal not opened in the OS folder; `*` used on Windows	`cd` into the correct folder; on Windows, enter each xml file name explicitly
Flashing stops midway	Loose cable, power loss, or connection through a USB hub	Power off, put the AIBOX back into EDL mode and flash again from the start; connect directly to the PC

Symptom	Common cause	Fix
AIBOX does not boot after flashing	Flashing incomplete, or the OS package was corrupted during download	Re-download and re-extract the OS package, then flash again from section 2

If the issue persists after the steps above, send the full output log of the `qdl` command to technical support.